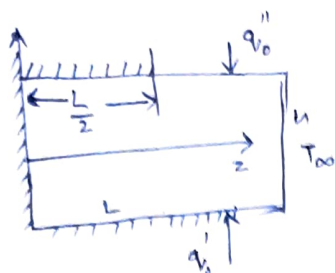


25 Jan 2024 D. Scott K. 09



$$\frac{1}{x} \cdot \frac{\partial}{\partial x} \left(kx \cdot \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial z} \left(k \cdot \frac{\partial T}{\partial z} \right) + q'' = 0$$

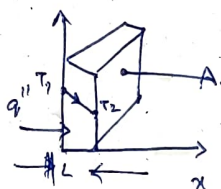
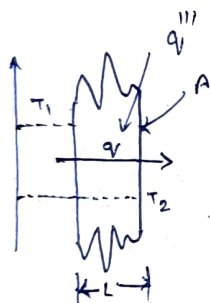
$$\frac{\partial T(0, x)}{\partial z} = 0$$

$$-k \cdot \frac{\partial T(L, x)}{\partial z} = h[T(L, x) - T_{\infty}]$$

$$\frac{\partial T(0, z)}{\partial x} = 0, \quad T(0, z) = \text{finite.}$$

$$k \cdot \frac{\partial T(x_0, z)}{\partial x} = f(z) = \begin{cases} 0 & 0 \leq z \leq L/2 \\ q''_0 & L/2 \leq z \leq L \end{cases}$$

Non-Dimensional form



Define:

- Reference length
- Reference temp
- Reference temp diff

$$\bar{T} \equiv \frac{T - T_1}{T_2 - T_1} \quad \bar{x} \equiv \frac{x}{L}$$

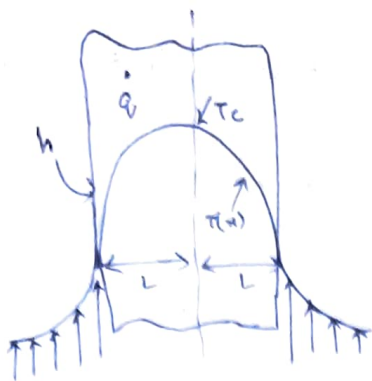
$$\frac{d^2 T}{dx^2} = -\frac{q''_0}{k}$$

$$\frac{d^2 \bar{T}}{d\bar{x}^2} = \frac{q''_0 \cdot L^2}{k(T_2 - T_1)} = -S$$

$$\bar{T}(\bar{x}=0) = 0 \quad \bar{T}(\bar{x}=1) = 1$$

$$\bar{T} = -\frac{S\bar{x}^2}{2} + C_1\bar{x} + C_2$$

$$\bar{T} = \bar{x} + \frac{S\bar{x}}{2} (1 - \bar{x})$$



Symmetric - solve half the problem

$$\frac{d^2 T}{dx^2} + \frac{\dot{q}}{k} = 0$$

$$\left. \frac{dT}{dx} \right|_{x=0} = 0$$

$$-k \left. \frac{dT}{dx} \right|_{x=L} = h \cdot (T(x=L) - T_{\infty})$$

Recall from Prev. problem

$$\frac{d^2 \bar{T}}{d\bar{x}^2} = - \frac{\dot{q}'' \cdot L^2}{16(T_2 - T_1)}$$

$$\bar{T} = \frac{(T - T_{\infty}) \cdot 16}{\dot{q}'' \cdot L^2}$$

Now

$$\frac{d^2 \bar{T}}{d\bar{x}^2} = -1$$

$$\left. \frac{d\bar{T}}{d\bar{x}} \right|_{\bar{x}=0} = 0, \quad \left. \frac{d\bar{T}}{d\bar{x}} \right|_{\bar{x}=1} = -Bi \bar{T}(\bar{x}=1)$$

Non-Dimensional form: 2D Heat eqⁿ

A stationary, homogeneous, isotropic solid is initially at a const. temp T_0 . For time $t > 0$, heat is generated within the solid and dissipated by convection from the bounding surfaces into a medium at const temp T_{∞} . Assume a rectangular geometry and a finite region, R .

BVP of heat conduction:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{k} = \frac{1}{\alpha} \cdot \frac{\partial T}{\partial t} \quad \text{in region } R, \quad t > 0$$

$$-k_i \cdot \frac{\partial T}{\partial \eta_i} = h_i (T_i - T_{\infty})$$

on s_i boundary of R ,
 $t > 0$

$i = 1, 2, \dots$ no. of continuous boundary surface of the solid.

$T = T_0$ in region R , $t = 0$

• Reference length $= L =$ a characteristic dimension of the solid.

• Define dimensionless length variables -

$$\xi = \frac{x}{L}, \quad \eta = \frac{y}{L}, \quad \psi = \frac{z}{L}$$

$\frac{\partial}{\partial N}$ = differentiation along outward-drawn normal in the new dimensionless coordinate system (ξ, η, ψ)

choose Ref. temp $= T_{\infty}$

$$\text{ref. temp dif.} = T_0 - T_{\infty}$$

Define dimensionless excess temp $\theta = \frac{T - T_{\infty}}{T_0 - T_{\infty}}$

GN substitution:

$$\frac{T_0 - T_{\infty}}{L^2} \cdot \left(\frac{\partial^2 \theta}{\partial \xi^2} + \frac{\partial^2 \theta}{\partial \eta^2} + \frac{\partial^2 \theta}{\partial \psi^2} \right) + \frac{\dot{q}}{L} = \frac{T_0 - T_{\infty}}{L^2} \cdot \frac{\partial \theta}{\partial t}$$

in $R, t > 0$

$$-\frac{k_i}{L} \cdot \frac{\partial \theta}{\partial N_i} = h_i \theta \quad \text{on } \partial_i, t > 0$$

$i=1, 2, \dots = \text{no. of continuous}$

$\theta = 1$ in $R, t = 0$

bounding surfaces.

$$\frac{\partial^2 \theta}{\partial \xi^2} + \frac{\partial^2 \theta}{\partial \eta^2} + \frac{\partial^2 \theta}{\partial \psi^2} + \frac{\dot{q} \cdot L^2}{(T_0 - T_{\infty}) \cdot k} = \frac{L^2}{\alpha} \cdot \frac{\partial \theta}{\partial t}$$

$$= \frac{\partial \theta}{\partial \tau} \quad \text{where } \tau = \frac{\alpha t}{L^2}$$

in $R, t > 0$

$$\Rightarrow \frac{\partial^2 \theta}{\partial \xi^2} + \frac{\partial^2 \theta}{\partial \eta^2} + \frac{\partial^2 \theta}{\partial \psi^2} + \phi = \frac{\partial \theta}{\partial \tau}$$

in $R, t > 0$

$$\frac{\partial \theta}{\partial n_i} = + \left(\frac{n_i L}{R_i} \right) \cdot \theta = 0$$

$$\Rightarrow \frac{\partial \theta}{\partial n_i} + B_i \theta = 0 \quad \text{on } S, \tau > 0$$

$$\theta = 1 \quad \text{in } R, \tau = 0$$

$$\tau = \frac{\alpha t}{L} = \frac{R t}{c_p L^2} = \frac{h L}{c_p L^2} = \frac{h L}{c_p L^2/t} = \left(\frac{h}{L} \right) \frac{L^2}{c_p L^2/t}$$

$$\phi = \frac{\dot{q} \cdot L^2}{(T_0 - T_\infty) \cdot R} = \frac{\dot{q} \cdot L^3}{\left(\frac{h}{L} \right) \cdot L^2 \cdot (T_0 - T_\infty)}$$

Solution of Homogenous Problem by separation of variables

$$\frac{1}{\alpha} \cdot \frac{\partial T(r, t)}{\partial t} = \nabla^2 \cdot T(r, t) \Rightarrow T(r, t) = \psi(r) \cdot \Gamma(t)$$

Boundary condition

$$\left[\begin{aligned} R_i \cdot \frac{\partial T}{\partial n_i} + h_i T(r, t) &= 0 \\ T(r, t=0) &= F(r) \end{aligned} \right] \Rightarrow \frac{1}{\alpha} \cdot \frac{\dot{\Gamma}(t)}{\Gamma(t)} = \frac{\nabla^2 \psi(r)}{\psi(r)} = -\lambda^2$$

Time

$$\dot{\Gamma}(t) + \alpha \lambda^2 \Gamma(t) = 0$$

$$\dot{\Gamma}(t) : e^{-\alpha \lambda^2 t}$$

Space

$$\nabla^2 \cdot \psi(r) + \lambda^2 \psi(r) = 0$$

$$R_i \cdot \frac{\partial \psi(r)}{\partial n_i} + h_i \psi(r) = 0$$

Helmholtz equation

$$\lambda = \lambda_m \longrightarrow \psi_m(r)$$

$$T(r, t) = \sum_{m=1}^{\infty} c_m \psi_m(r) \cdot e^{-\alpha \lambda_m^2 t}$$

Eigenvectors

$$c_m = \frac{\int_R \psi_m(r) \cdot F(r) \cdot dV}{\int_R \psi_m^2(r) \cdot dV}$$

Norm.

Normalized eigen function: $k = \frac{\psi_m(r)}{\sqrt{N}}$

Uniqueness of solution for 3D Heat Equation

that
Prove the solution of the following 3D-Heat problem is unique,

$$\begin{aligned} u_t &= \nabla^2 u, & x \in D & \quad (\text{false } u=T) \\ u(x, t) &= 0, & x \in \partial D & \text{ on the boundary.} \\ u(x, 0) &= f(x), & x \in D \end{aligned}$$

Let, u_1 and u_2 be two solⁿ. Define $v = u_1 - u_2$ then v satisfies

$$v_t = \nabla^2 v, \quad x \in D$$

$$\begin{aligned} v(x, t) &= 0, & x \in \partial D \\ v(x, 0) &= 0, & x \in D. \end{aligned}$$

Define

$$V(t) = \iiint_D v^2 dV \geq 0$$

$V(t) \geq 0$ since the integrand $v^2(x, t) \geq 0$ for all (x, t) .

$$V(t) = \iiint_D v^2 dV \geq 0 \quad v_t = \nabla^2 v$$

$$\frac{dV}{dt}(t) = \iiint_D 2v v_t dV$$

substituting for v_t
from the PDE
yields

$$\frac{dV}{dt}(t) = \iiint_D 2v \cdot \nabla^2 v dV$$

Now use,

$$\iint_S v \nabla v \cdot \hat{n} dS = \iiint_V (v \nabla \cdot \nabla v + \nabla v \cdot \nabla v) dV$$

$$= \iiint_V (v \cdot \nabla^2 v + |\nabla v|^2) dV$$

we get.

$$\frac{dV}{dt}(t) = 2 \cdot \iint_{\partial D} v \nabla v \cdot \hat{n} dS - 2 \iiint_D |\nabla v|^2 dV$$

$$\frac{dV}{dt}(t) = 2 \int_{\partial D} \mathbf{v} \cdot \nabla \mathbf{v} \cdot \hat{\mathbf{n}} \cdot dS - 2 \iiint_D |\nabla \mathbf{v}|^2 dV$$

But on ∂D , $\mathbf{v} = 0$, so that the 1st integral on the RHS vanishes.

Thus,

$$\frac{dV}{dt}(t) = -2 \iiint_D |\nabla \mathbf{v}|^2 \cdot dV \leq 0$$

At $t = 0$:

$$V(0) = \iiint_D v^2(x, 0) \cdot dV = 0$$

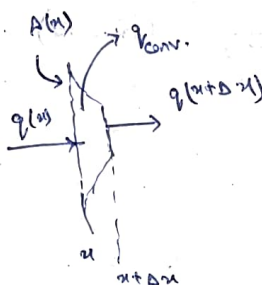
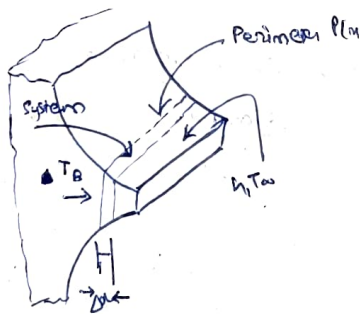
Recall: $v(x, 0) = 0$

Thus, $V(0) = 0$, $V(t)$ and $dV/dt \leq 0$

$V(t)$ is a non-negative, non-increasing function that starts at zero.

Thus $V(t)$ must be zero for all time t , so that $v(x, t)$ must

Heat transfer in Extended Surfaces: Fins, Spine



energy balance:

$$q(x) = q(x+dx) + q_{conv}$$

$$\downarrow$$

$$= q(x) + \frac{dq}{dx} \cdot dx + q_{conv}$$

Also,

$$q_{conv} = h \cdot P(x) \cdot dx \cdot (T - T_\infty)$$

$$\frac{dq}{dx} + h \cdot P(x) \cdot (T - T_{\infty}) = 0$$

Use: $q(x) = -k \cdot A(x) \cdot \frac{dT}{dx}$

$$\frac{d}{dx} \left[A(x) \cdot \frac{dT}{dx} \right] - h \frac{P(x)}{k} \cdot (T - T_{\infty}) = 0 \quad \text{Fin Equation.}$$

$$\bar{T} = \frac{T - T_{\infty}}{T_b - T_{\infty}}, \quad \bar{x} = \frac{x}{L}$$

$$\frac{d}{d\bar{x}} \cdot A_c \cdot \frac{d\bar{T}}{d\bar{x}} - \frac{h \cdot P \cdot L^2}{k} \cdot \bar{T} = 0$$

→ It is non-dimensional, each term has unit area.

One boundary condition is at Fin Base: $T = T_b, \bar{T}(\bar{x}=0) = 1$

Three types of BC at fin tip:

fixed temp $\leftarrow T(\bar{x}=1) = \bar{T}_t$

→ known T at fin tip

insulated tip $\leftarrow \frac{d\bar{T}}{d\bar{x}} \Big|_{\bar{x}=1} = 0$

→ Adiabatic fin tip

$-\frac{d\bar{T}}{d\bar{x}} \Big|_{\bar{x}=1} = Bi_t, \quad \bar{T}(\bar{x}=1) \rightarrow$ Convection at fin tip.

why ID?  Fin Equation

If y denotes the direction normal to the surface area, the energy balance at surface would be

$$-k \cdot \frac{\partial T}{\partial y} \Big|_{y=b} = h(T - T_{\infty})$$

approximate the derivative as:

$$\left. \frac{\partial T}{\partial y} \right|_{y=b} \approx \frac{\Delta T}{b} \quad , \quad \Delta T \text{ is avg. temp difference across the fin in } y\text{-direction.}$$

$$K \cdot \frac{\Delta T}{b} = h(T - T_{\infty})$$

$$\Rightarrow \Delta T = \frac{hb}{K} \cdot (T - T_{\infty})$$

$$\underbrace{\frac{\Delta T}{(T_B - T_a)}}_{\Delta \bar{T}} = \frac{hb}{K} \cdot \underbrace{\left(\frac{T - T_{\infty}}{T_B - T_{\infty}} \right)}_{\bar{T}}$$

$$\Delta \bar{T} \approx \frac{h \cdot b}{K} \cdot \bar{T} = Bi_b \cdot \bar{T}$$

For 1D assumption to be correct we ^{expect} $\Delta \bar{T} \ll \bar{T}$

i.e., variation in temp in y -Direction is much smaller than variation in x -Direction

$\therefore$ 1D assumption is correct

Now,

$$\frac{d}{d\bar{x}} \cdot A_c \frac{d\bar{T}}{d\bar{x}} - \frac{h \cdot P \cdot L^2}{K} \cdot \bar{T} = 0$$

If A_c is const:

$$\bar{T}'' - N^2 \cdot \bar{T} = 0$$

$$\Rightarrow N^2 = \frac{h \cdot P L^2}{K A_c}$$

$$\text{Solution: } \bar{T} = A \cdot e^{N\bar{x}} + B \cdot e^{-N\bar{x}}$$

For adiabatic fin tip:

$$\bar{T} = \frac{e^{N(1-\bar{x})} + e^{-N(1-\bar{x})}}{e^N + e^{-N}}$$

$$A = \frac{e^{-N}}{e^N + e^{-N}}$$

$$B = 1 - A = \frac{e^N}{e^N + e^{-N}}$$

$$= \frac{\cosh[N(1-\bar{x})]}{\cosh(N)}$$

All the heat removed from the fin must be transported into the fin at the base by conduction. This gives

$$q = -k \cdot A_{c,b} \cdot \left. \frac{dT}{dx} \right|_{x=0} = \frac{k \cdot A_{c,b} \cdot (T_B - T_\infty)}{L} \cdot \left. \frac{dT}{dx} \right|_{x=0}$$

$$q = \frac{k A_c (T_B - T_\infty)}{L} \cdot N \cdot \tanh(N) = \sqrt{h P k A_c} (T_B - T_\infty) \cdot \tanh(N)$$

Now, for a given cross-section area and environment where fin needs to be used we can increase the heat removed from the fin by increasing N .

i.e. longer fin will transfer more heat.

$$N^2 = \frac{h \cdot P \cdot L^2}{k \cdot A_c}$$

$$\tanh(N) \rightarrow 1 \text{ for } N \gg 1$$

⊙ Longer the fin - Higher is the heat removal.

$$\tanh(3) = 0.995$$

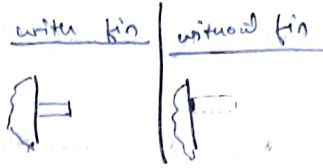
∴ for $N > 3$ is essentially infinite in length.

For design viewpoint, Rule: $N > 2$ to 2.5 ,

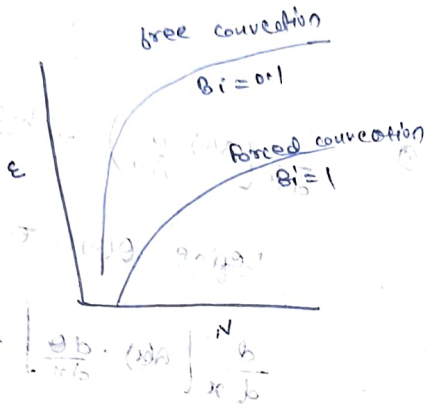
Performance Measurement : Fin effectiveness

$$\epsilon = \frac{q_{\text{fin}}}{q_{\text{without fin}}} = \frac{\sqrt{h \cdot P \cdot k \cdot A_c} (T_b - T_{\infty}) \cdot \tanh h(N)}{h \cdot A_c \cdot (T_b - T_{\infty})}$$

$$= \frac{N}{Bi} \tanh h(N)$$



⊛ Fin is justified if $\epsilon > 2$



⇒ fin is more effective for free conv. or lower Bi

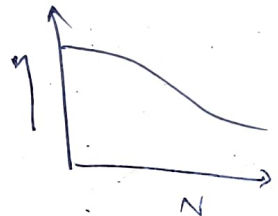
Performance measurement: Efficiency

$$q = \frac{k \cdot A_c \cdot (T_b - T_{\infty})}{L} \cdot N \cdot \tanh h(N)$$

efficiency is defined as Ratio of the actual to theoretical maximum heat transfer from the fin.

$$q_{\text{max}} = h \cdot A_{\text{fin}} \cdot (T_b - T_{\infty}) = h \cdot P \cdot (T_b - T_{\infty}), \text{ uniform } A_c$$

$$\eta = \frac{\tanh h(N)}{N}$$



Fin equation : Use $\Theta = T - T_{\infty}$

$$\frac{d}{dx} \cdot \left[A(x) \cdot \frac{dT}{dx} \right] - \frac{h \cdot P(x) \cdot (T - T_{\infty})}{k} = 0$$

Defining new temp function

$$\Theta(x) = T(x) - T_{\infty}$$

$$\frac{d}{dx} \left[A(x) \cdot \frac{d\theta}{dx} \right] - \frac{h \cdot P(x)}{k} \theta = 0$$

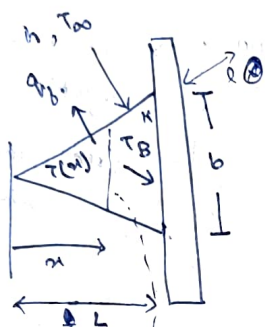
Solution:

$$\theta(x) = c_1 \cdot e^{-mx} + c_2 \cdot e^{mx}$$

Fins: non-uniform cross section: Triangular fin

Fins of non-uniform cross section can usually transfer more heat for a given mass more than those of a const. cross section.

Cross section.



width of
fin = b

$$\frac{d}{dx} \left[A(x) \cdot \frac{dT}{dx} \right] - \frac{h \cdot P(x)}{k} (T - T_\infty) = 0$$

Define $\theta(x) = T(x) - T_\infty$

$$\frac{d}{dx} \left[A(x) \cdot \frac{d\theta}{dx} \right] - \frac{h \cdot P(x)}{k} \theta = 0$$

$$A(x) = \left(\frac{bx}{L} \right) L$$

and,

$$P(x) = 2 \cdot \left(\frac{bx}{L} + L \right)$$

$$\left\{ \begin{array}{l} \frac{b}{L} = \left(\frac{1}{x} \right) \\ \Rightarrow \left(\frac{1}{x} \right) = \frac{b \cdot \ln}{L} \end{array} \right\}$$

If we assume $b \ll L$, then $P(x) \approx 2L$

$$\frac{d}{dx} \left(x \cdot \frac{d\theta}{dx} \right) - m^2 \cdot x \cdot \theta = 0$$

where, $m^2 = \frac{2h \cdot L}{k_b}$

$$x^2 \cdot \frac{d^2\theta}{dx^2} + x \cdot \frac{d\theta}{dx} - m^2 \cdot x \cdot \theta = 0$$

multiply both sides by x .

Define $\eta = \sqrt{x}$.

$$\eta^2 \cdot \frac{d^2 \theta}{d\eta^2} + \eta \cdot \frac{d\theta}{d\eta} - 4m^2 \cdot \eta^2 \cdot \theta = 0$$

Modified
Bessel equation

Solution:

$$\theta(\eta) = C_1 \cdot I_0(2m\eta) + C_2 \cdot K_0(2m\eta)$$

$$\theta(x) = C_1 \cdot I_0(2m\sqrt{x}) + C_2 \cdot K_0(2m\sqrt{x})$$

$$T(0) = \text{finite} \Rightarrow \theta(0) = \text{finite}$$

$$T(L) = T_b \Rightarrow \theta(L) = T_b - T_\infty = \theta_b$$

$C_2 = 0$ else K_0 will be infinite.

$$K_0(0) \rightarrow \infty, \quad C_2 = 0$$

$$C_1 = \frac{\theta_b}{I_0(2m\sqrt{L})}$$

$$\frac{\theta(x)}{\theta_b} = \frac{T(x) - T_\infty}{T_b - T_\infty} = \frac{I_0(2m\sqrt{x})}{I_0(2m\sqrt{L})}$$

Triangular fins: Rate of Heat transfer

$$\frac{\theta(x)}{\theta_b} = \frac{T(x) - T_b}{T_b - T_\infty} = \frac{I_0(2m\sqrt{x})}{I_0(2m\sqrt{L})}$$

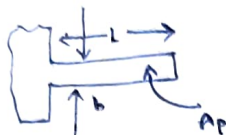
$$q_b = k \cdot A \left(\frac{dT}{dx} \right)_{x=L} = k \cdot A \left(\frac{d\theta}{dx} \right)_{x=L}$$

$$\frac{d}{dx} [I_0(\alpha x)] = \alpha \cdot I_1(\alpha x)$$

$$q_b = l \cdot \sqrt{2h \cdot k_b} \cdot \theta_b \cdot \frac{I_1(2m\sqrt{L})}{I_0(2m\sqrt{L})}$$

Fin optimization: Rectangular fin

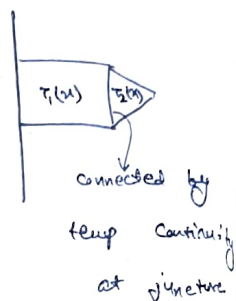
For a given fin shape, fin material, and convection conditions, there exists an optimized design which transfers the maximum amount of heat for a given mass of the fin.



consider: Adiabatic fin tip

$$q = \sqrt{h \cdot P k (A_c)} \cdot (T_b - T_\infty) \cdot \tanh(N)$$

$$N^2 = \frac{h \cdot P L^2}{k A_c}$$



For a long fin ($w \gg b$), $P \approx 2w$ and $A_c = b \cdot w$, Thus:

$$q' = \frac{q}{w} = \sqrt{2b h k} \cdot (T_b - T_\infty) \tanh(N)$$

$$N^2 = \frac{2hL^2}{kb}$$

The length L can be eliminated using $A_p = b \cdot L$. The formula for

$$N \text{ becomes } N^2 = \frac{2 \cdot h \cdot A_p^2}{k \cdot b^3} \Rightarrow b = \left(\frac{2 \cdot h \cdot A_p^2}{k N^2} \right)^{1/3}$$

$$q' = \underbrace{\left(4 h^2 \cdot k A_p \right)^{1/3} \cdot (T_b - T_\infty)^{2/3}}_{\text{function of } N} \cdot N^{1/3} \cdot \tanh(N)$$

$$f(N) = N^{1/3} \cdot \tanh(N) \quad \text{set } \frac{df}{dN} = 0$$

$$\Rightarrow \cos h(N) \cdot \sinh(N) - 3N = 0$$

solve for N

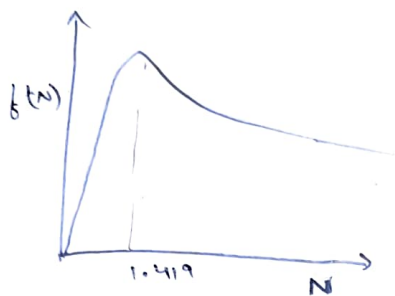
$$N_{opt} = 1.419 = \left(\frac{2 \cdot h \cdot A_p^2}{k \cdot b_{opt}^3} \right)^{1/2}$$

once A_p is fixed, use the above formula to obtain b_{opt} ,
and find L from $L = \frac{A_p}{b_{opt}}$

Rate of heat transfer : $q'_{opt} = \left(\frac{4h^2 \cdot k \cdot A_p}{N_{opt}} \right)^{1/3} \cdot (T_b - T_\infty) \cdot \tanh(N_{opt})$

$$= 1.256 (h^2 \cdot k \cdot A_p)^{1/3} \cdot (T_b - T_\infty)$$

$$f(N) = N^{-Y_3} \cdot \tan h(N)$$



$$N_{opt} = 1.419 = \left(\frac{2h \cdot A^2 p}{k b_{opt}^3} \right)^{1/2}$$

$$q'_{opt} = \left(\frac{4h^2 \cdot K_{AP}}{N_{opt}} \right)^{1/3} \cdot \frac{(T_b - T_{\infty})}{\tanh h(N_{opt})}$$

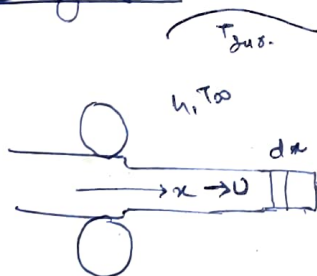
$$= 1.256 \cdot (h^2 K_{AP})^{1/3} (T_b - T_{\infty})$$

for Triangular fin:

$$q'_{opt} = 1.422 \cdot (h^2 \cdot k \cdot A_p)^{1/3} \cdot (T_b - T_\infty)$$

which one is better - Rectangular fin or triangular fin?
→ Triangular fin.

Moving fin



Examples

- > extrusion of plastics
- > drawing of wires and sheet.

- > sheet been drawn with velocity U through rollers.
- > Temperature varies with distance from rollers.

Find $\pi(x)$.

- > Sheet exchanges heat with surrounding by radiation
→ " " " " " " " " " " " "
-
-